

Final Report

Introduction

Financial markets are among the most complex and dynamic systems studied within data science. Stock prices are influenced simultaneously by:

- Macroeconomic conditions
- Investor psychology
- Exchange-rate movements
- Monetary policy decisions
- Global market sentiment
- Political uncertainty
- External economic shocks

Unlike deterministic systems, financial markets behave stochastically, meaning that uncertainty and unpredictability are fundamental characteristics of price behavior.

Because relationships between financial variables continuously evolve over time, predicting stock returns remains one of the most difficult applications of statistical analysis and machine learning.

The Turkish stock market provides a particularly interesting environment for financial analysis because emerging markets are generally more sensitive to:

- Inflation
- Exchange-rate volatility
- Macroeconomic instability
- Political uncertainty

The primary purpose of this project is to investigate whether statistical analysis and machine learning techniques can identify meaningful relationships between Turkish stocks and broader market conditions represented by the BIST100 index.

Rather than focusing solely on stock prices, the project aims to understand:

- Market dependency
- Volatility behavior
- Tail risk
- Predictive limitations
- Financial uncertainty

through a combination of:

- Exploratory Data Analysis (EDA)
- Statistical inference

- Regression analysis
- Machine learning classification models

An additional objective of the project is to examine the distinction between:

- Correlation
- Predictability

because strong financial correlations do not necessarily imply reliable forecasting power.

Research Questions

The project investigates four primary research questions:

- Do Turkish large-cap stocks systematically move together with the BIST100 market index?
- How do volatility, rolling beta values, and drawdowns evolve under changing market conditions?
- Are stock returns statistically different from overall market returns?
- Can machine learning algorithms predict the next-day direction of stock returns using historical financial indicators?

The ultimate goal is not simply to maximize prediction accuracy, but to better understand the structure and limitations of Turkish financial markets.

Dataset and Data Collection

Historical daily financial data was collected using:

- Yahoo Finance API
- yfinance Python library

The dataset covers the period:

- 2018–2025

This period includes several economically important events:

- Post-pandemic recovery
- Inflationary periods
- Exchange-rate instability
- Monetary policy changes
- Increased market volatility

The selected assets include:

- THYAO.IS
- GARAN.IS

- ASELS.IS
- KCHOL.IS
- TUPRS.IS
- BIMAS.IS
- BIST100 Index (^XU100)
- S&P500 Index (^GSPC)
- USD/TRY Exchange Rate (USDTRY=X)

For each asset, the following variables were collected:

- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume

Adjusted closing prices were used for return calculations because they account for dividends and stock splits.

Data Preprocessing

Several preprocessing operations were required before analysis:

- Date alignment
- Missing value handling
- Forward filling
- Infinite value cleaning
- Return calculations
- Feature engineering

Daily percentage returns were calculated instead of raw prices to:

- Normalize performance
- Improve comparability
- Reduce scale dependency

Additional engineered features include:

- Lagged returns
- Rolling averages
- Rolling volatility
- Momentum indicators
- Market return variables

These engineered variables were designed to improve both statistical interpretation and predictive performance.

Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand:

- Return behavior
- Market dependency
- Volatility persistence
- Correlation structures
- Downside risk
- Portfolio sensitivity

Return Distribution Analysis

The selected Turkish stocks exhibited significant differences in:

- Return behavior
- Volatility
- Drawdown exposure

ASELS generated the highest annualized return among the selected assets but also displayed elevated volatility levels.

THYAO demonstrated strong returns together with the largest maximum drawdown, indicating high exposure to macroeconomic instability and market shocks.

BIMAS showed relatively lower volatility and smaller drawdowns, suggesting more defensive behavior during periods of economic stress.

Return distributions were found to be:

- Non-normal
- Heavy-tailed
- Asymmetric

This finding is important because many traditional financial models assume normally distributed returns.

Cumulative Return Analysis

Cumulative return analysis revealed that the equal-weighted portfolio closely followed the trajectory of the BIST100 index.

This suggests that:

- Systematic market risk dominates firm-specific effects
- Turkish large-cap stocks move together under similar market conditions
- Local market behavior strongly influences portfolio returns

Correlation Analysis

A correlation heatmap was used to examine relationships between:

- Turkish stocks
- Portfolio returns
- BIST100 returns
- USD/TRY exchange rate
- S&P500 index

The analysis revealed:

- Strong positive correlations between most stocks and BIST100
- Significant market dependence among Turkish large-cap equities
- Lower correlation values for BIMAS compared to other assets

THYAO and GARAN displayed especially strong market sensitivity, indicating that:

- Banking and aviation sectors are highly affected by macroeconomic conditions
- Systematic market risk dominates sector behavior

However, one of the most important findings is that:

Correlation does not necessarily imply predictability.

Although stocks move together under similar conditions, these relationships may still fail to produce reliable forecasts for future returns.

Volatility Analysis

Rolling volatility analysis revealed:

- Volatility clustering
- Persistent financial uncertainty
- Dynamic market behavior

Periods of high volatility were frequently followed by additional high-volatility periods.

THYAO and GARAN experienced especially large volatility spikes during unstable periods, while BIMAS exhibited smoother volatility behavior.

This finding confirms that:

- Financial risk changes over time
- Market shocks create persistent uncertainty
- Investor behavior evolves dynamically under stress

Rolling Beta Analysis

Rolling beta analysis was conducted to evaluate changing market sensitivity over time.

The analysis showed that:

- Beta values fluctuate significantly
- Market dependency increases during unstable periods
- Investors react more strongly to market-wide developments during periods of uncertainty

This confirms that systematic market risk becomes dominant during periods of financial stress.

Drawdown Analysis

Maximum drawdown analysis revealed that:

- THYAO experienced the largest drawdown
- BIMAS displayed lower downside exposure
- High-return assets often involve substantial short-term losses

This highlights the importance of:

- Risk management
- Portfolio stability
- Investor psychology

Volume and Volatility Relationship

The relationship between trading volume and volatility was examined using scatter plots comparing:

- Average trading volume
- Absolute return magnitude

The analysis showed a positive relationship between:

- Trading activity
- Volatility

Higher trading volumes were generally associated with larger price movements.

This result is consistent with financial theory, where increased trading activity often reflects heightened uncertainty and information flow.

Hypothesis Testing and Statistical Inference

Several statistical tests were conducted to evaluate whether observed relationships were statistically significant.

One-Sample t-Test

The one-sample t-test examined whether mean stock returns differed significantly from zero.

Most p-values exceeded the 0.05 significance level, meaning that the null hypothesis could not be rejected for most assets.

This finding supports the weak form of market efficiency.

Paired t-Test

A paired t-test compared portfolio returns with BIST100 returns.

Results showed that:

- The portfolio did not significantly outperform the market
- Market-wide dynamics dominate stock behavior

This further confirms the strong dependency between Turkish large-cap stocks and the broader market.

Pearson Correlation Test

Pearson correlation tests examined the relationship between:

- Trading volume
- Volatility

The results were statistically significant, allowing rejection of the null hypothesis.

This confirms that:

- Higher trading activity is associated with increased volatility
- Market uncertainty intensifies during periods of elevated volume

Jarque-Bera Test

Jarque-Bera tests revealed:

- Heavy-tailed distributions
- Non-normal return behavior
- Presence of extreme observations

This result is important because many classical financial models assume normality.

Augmented Dickey-Fuller Test

ADF tests confirmed stationarity of daily return series.

Stationarity is essential because:

- Non-stationary data may generate misleading statistical relationships
- Stable statistical properties improve regression and machine learning reliability

Regression Analysis

Regression analysis was performed to quantify relationships between:

- Portfolio returns
- BIST100 returns
- USD/TRY returns
- S&P500 returns

The results demonstrated that:

- BIST100 returns possess the strongest explanatory power
- Local market conditions dominate Turkish stock behavior
- S&P500 exhibits moderate influence
- USD/TRY shows weaker but still meaningful effects

These findings confirm that:

- Systematic market exposure dominates firm-specific effects
- Turkish stocks remain highly dependent on domestic market conditions

Machine Learning Analysis

The machine learning stage evaluated whether relationships identified during EDA could be transformed into meaningful predictive signals.

The prediction problem was formulated as a binary classification task:

- Positive next-day return
- Negative next-day return

The dataset was divided into:

- Training set: 1239 observations
- Test set: 310 observations

A time-based split was used to avoid look-ahead bias.

Features Used

The machine learning models used:

- BIST100 returns
- Lagged portfolio returns
- Rolling volatility
- Short-term rolling trend indicators
- Market-related variables

These features were selected based on:

- Volatility clustering
- Heavy-tailed behavior
- Strong market dependence

identified during the EDA stage.

Machine Learning Models

The project implemented:

- Logistic Regression
- Random Forest
- Gradient Boosting

Each model represents different levels of complexity and non-linearity.

Logistic Regression Results

Logistic Regression achieved:

- Accuracy ≈ 0.50
- ROC-AUC ≈ 0.494

This indicates performance close to random guessing.

The result suggests that the relationship between selected features and next-day return direction is not linearly separable.

Random Forest Results

Random Forest achieved:

- Accuracy ≈ 0.516
- ROC-AUC ≈ 0.504

Although this model produced the best performance, the improvement remained marginal.

The low ROC-AUC value indicates that predictive power remains weak even when non-linear relationships are considered.

Gradient Boosting Results

Gradient Boosting achieved:

- Accuracy ≈ 0.464
- ROC-AUC ≈ 0.472

This model performed worse than the others, suggesting that it may overfit noise within financial return data.

Integrated Machine Learning Interpretation

The machine learning analysis demonstrates that:

- Models can identify weak relationships
- Predictive signals remain extremely limited
- Financial return behavior is highly noisy and uncertain

A key insight emerges when machine learning results are compared with EDA findings.

The EDA stage revealed:

- Strong market dependence
- Significant correlations
- Systematic market exposure

However, these relationships do not translate into strong predictive performance.

This means that:

- Stocks may move together under similar conditions
- But historical information alone cannot reliably predict future return direction

The heavy-tailed and volatile nature of returns further reduces model reliability, especially during unstable periods.

Overall Interpretation

The project demonstrates several important findings:

- Turkish large-cap stocks are strongly influenced by BIST100 market behavior
- Financial returns exhibit heavy tails and volatility clustering
- Market-wide dynamics dominate portfolio performance
- Strong correlation does not guarantee predictability
- Machine learning models provide limited short-term forecasting power

The findings support the idea that financial markets are highly efficient in the short term.

Historical financial information alone contains limited predictive content for future return direction.

As a result, machine learning models in this context are more useful for:

- Understanding market structure
- Interpreting financial behavior
- Evaluating predictive limitations

rather than producing highly accurate forecasts.

Technologies Used

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- yfinance

Platforms

- Google Colab
- GitHub

AI and Development Tools Used

The project development process included the use of AI-assisted and development tools to improve:

- Documentation quality
- Code organization
- Debugging
- Interpretation support

Tools used include:

- ChatGPT
- Google Colab
- GitHub

All analytical decisions, statistical evaluations, preprocessing stages, and machine learning implementations were reviewed and validated throughout the project development process.

Conclusion

This project combines:

- Financial analysis
- Exploratory data analysis
- Statistical inference
- Regression modeling
- Machine learning classification

within a single analytical framework to investigate Turkish stock market behavior.

The findings highlight the complex, dynamic, and uncertain nature of financial markets.

Although strong relationships between stocks and market conditions exist, these relationships do not necessarily produce meaningful predictive power.

Overall, the project demonstrates both the capabilities and the limitations of machine learning techniques in financial market prediction.